

When Better APT Scores Hide Missed Attack Warnings

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A better score can hide a lost warning

- A wrong attack stage can still trigger review.
- A benign prediction can suppress that warning.
- Macro-F1 alone does not describe this difference.
- Warning recall must be read with benign false alerts.

Problem, purpose and thesis

- Problem: score improvements can conceal missed stage warnings.
- Purpose: expose the tradeoff on the same evaluation records.
- Thesis: review score, warning loss and false alerts together.
- A valid comparison also needs credible labels and chronology.

Three research questions

- RQ1: When score improves, what happens to warnings and false alerts?
- RQ2: How much does later-period training access change a fixed-anchor score?
- RQ3: Which proposed releases support the unchanged chronological task?
- Warning-loss analysis is retrospective; it is not preregistered.

The applied contribution

- Paired evidence of score gains with lost exfiltration warnings.
- A fixed-anchor training-composition contrast.
- An executable source-qualification and model-review record.
- Uncapped coverage and a separate AIT execution test.

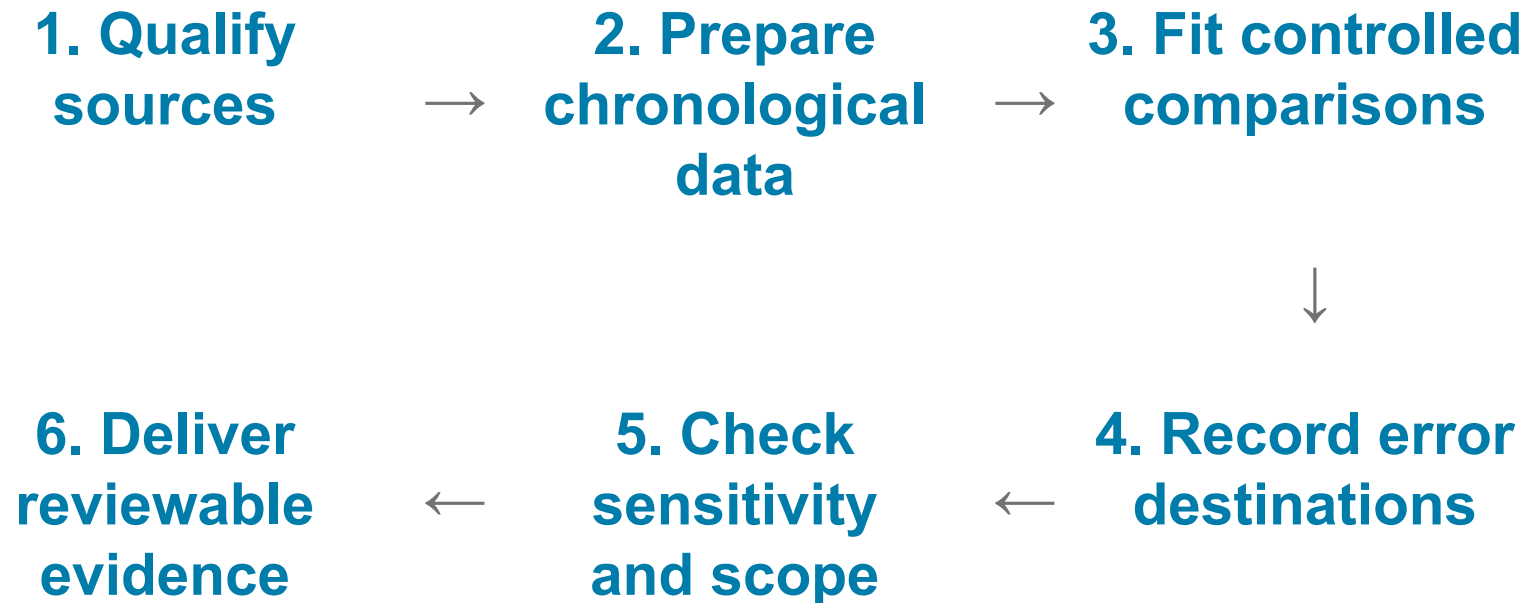
What the closest work already establishes

- TESSERACT: temporal and distribution constraints.
- Bilot / Guerra: practical and benchmark validity in APT evaluation.
- Uddin: wrong attack type versus attack called normal.
- This study: a controlled flow-stage tradeoff and traceable audit.

The evidence has four distinct scopes

- Original controlled study: 382,229 source rows.
- UNRAVELED full release: 6,877,157 rows; same campaign.
- AIT: eight executions; two later runs held out.
- Source qualification retains other releases' eligibility limits.

Graphical model of the research



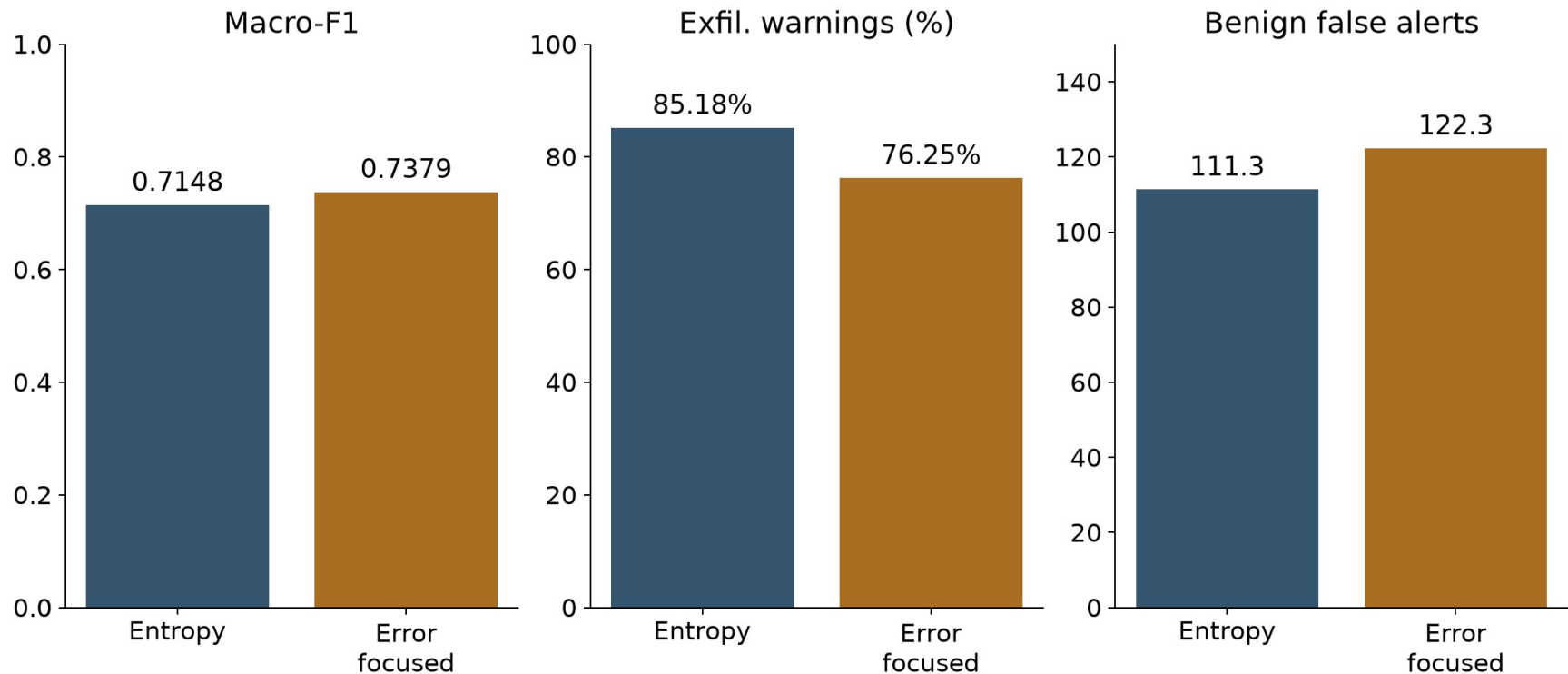
Controls for a valid comparison

- Compare the same evaluation rows within each paired contrast.
- Keep architecture and class budgets fixed in the temporal contrast.
- Freeze decisions before evaluating later test observations.
- Retain adverse conditions, seed variation and source limitations.

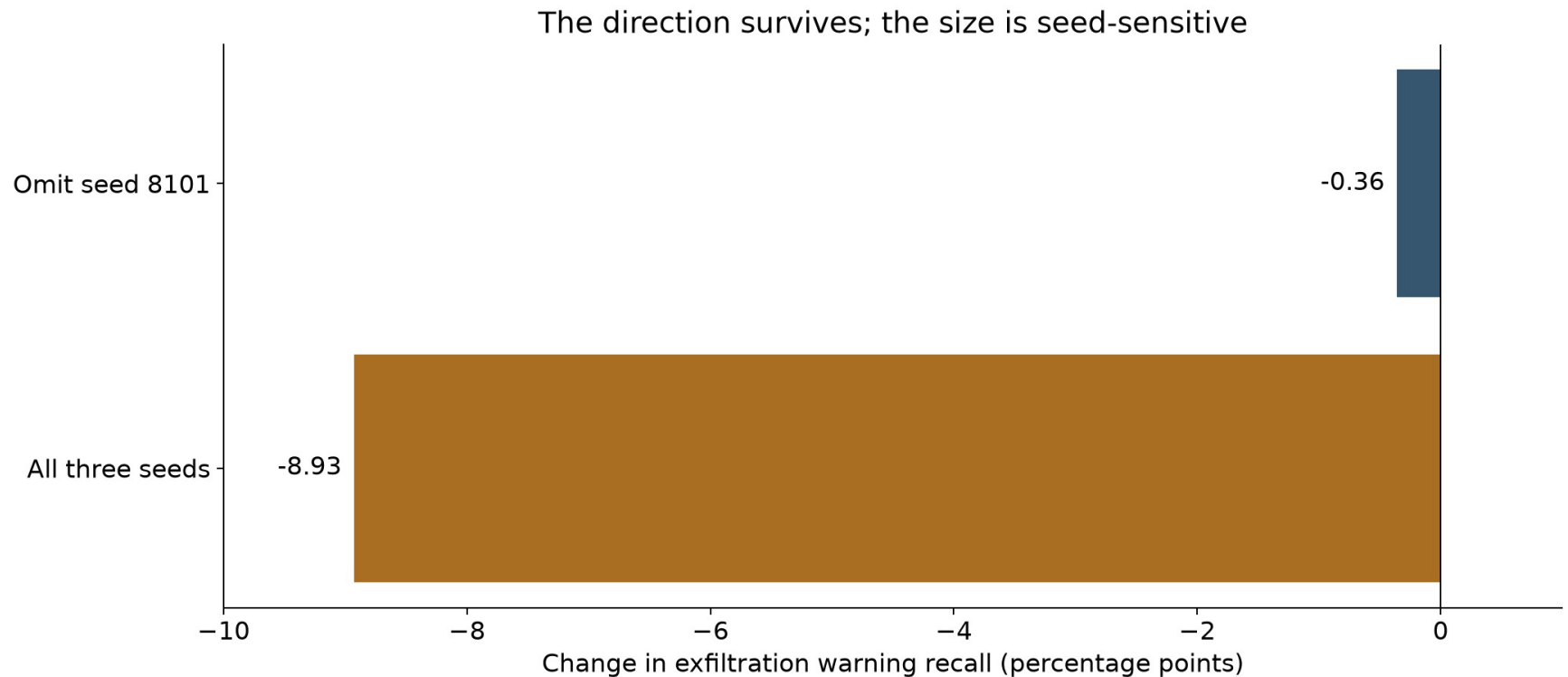
Read the score and the error destination

- Macro-F1: average of declared class F1 scores.
- Exact-stage recall: correct stage / true stage records.
- Warning recall: any attack prediction / true stage records.
- False-alert rate: benign records called attack / benign records.

Higher F1, fewer exfiltration warnings



The warning-loss size is seed-sensitive



Training composition changes the score

- The evaluation anchor stays fixed at 104,051 rows.
- Architecture and per-class fitting counts stay fixed.
- Allowing later observations increases mean macro-F1 by 0.0632.
- The contrast measures composition/access sensitivity, not pure time causality.

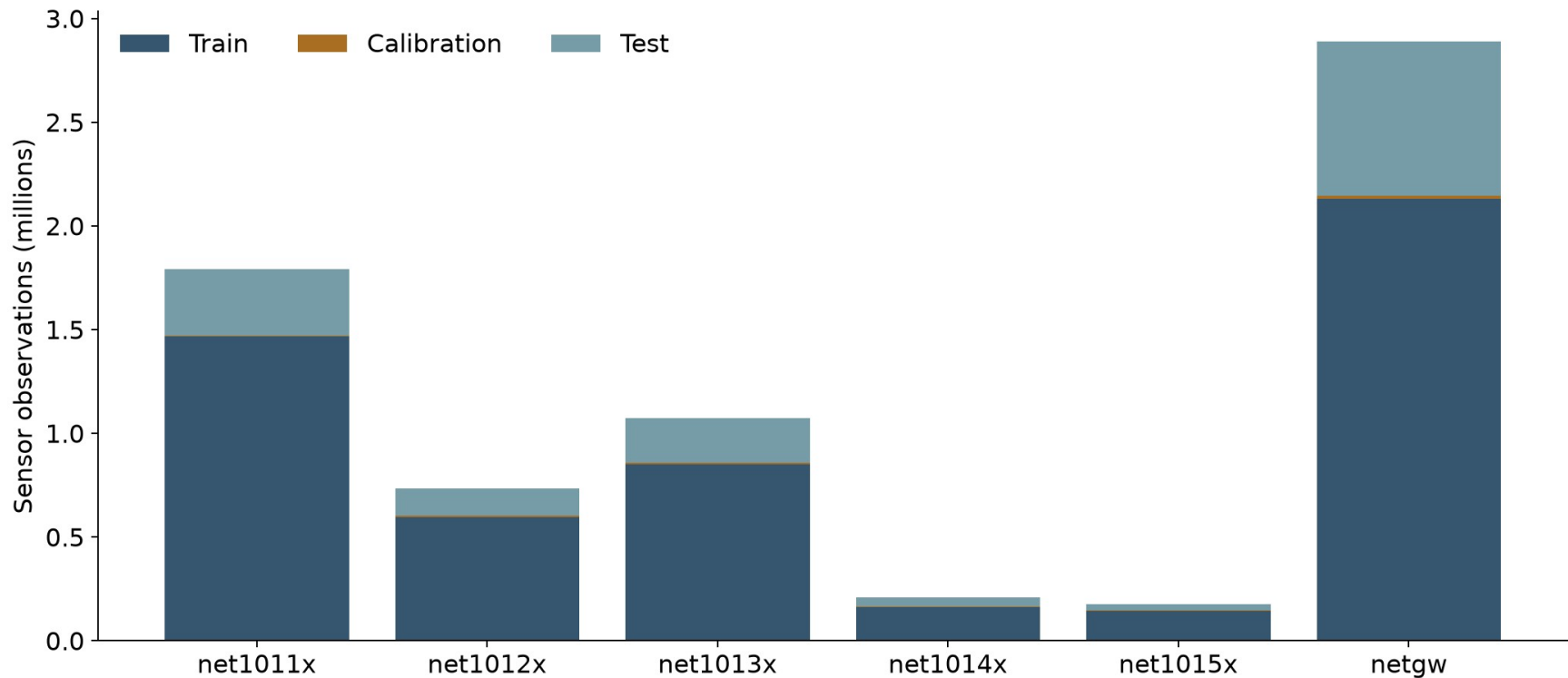
History can help and still lose warnings

- Chronological macro-F1: 0.7365 $\rightarrow$ 0.7582.
- Mean benign false alerts: 24.0 $\rightarrow$ 14.3.
- A smaller stage-warning loss remains visible.
- The joint report exposes benefits and costs together.

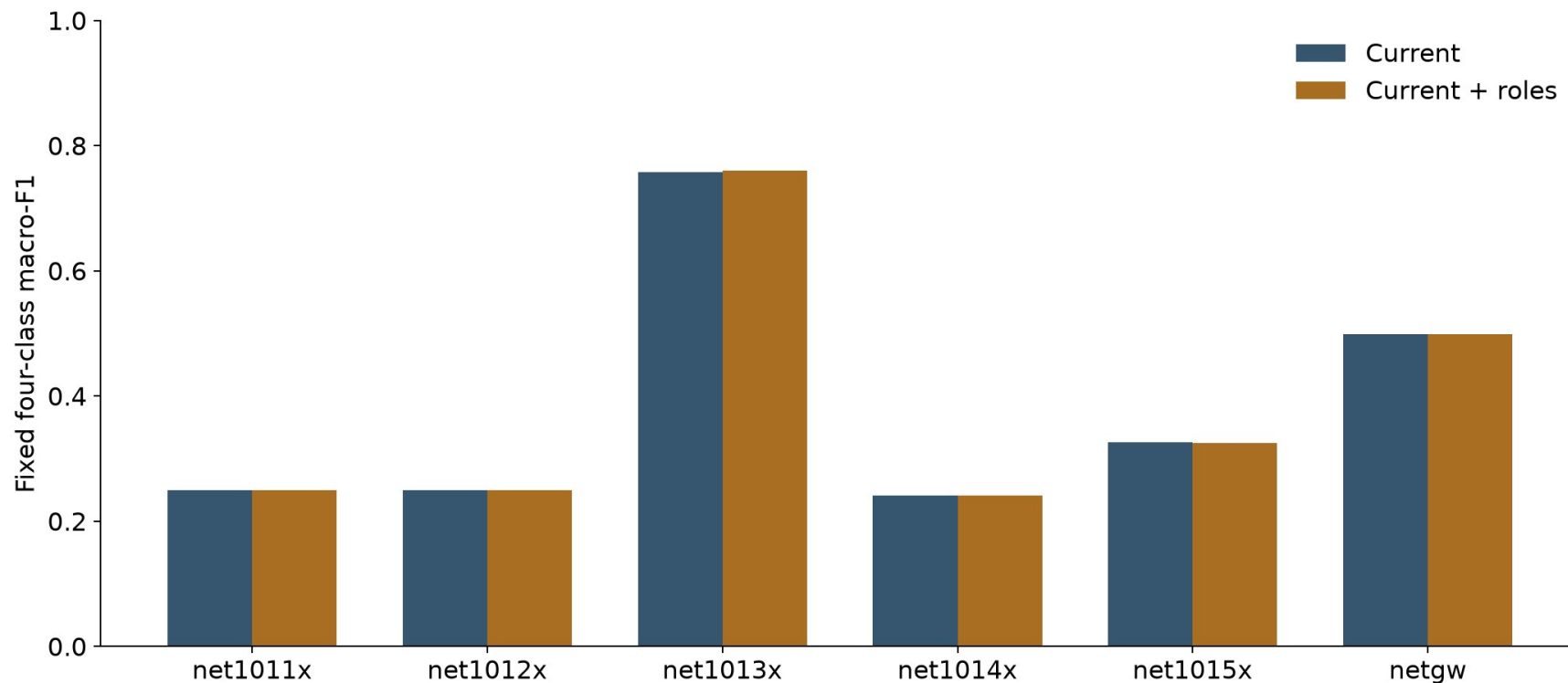
Full release: frozen before new fitting

- All 173 files verified against source hashes and native counts.
- Train: before June 27; calibrate: June 27; test: June 28 onward, UTC.
- Every eligible training row; current versus current + roles.
- Duplicates, label conflicts and cutoff-crossing flows accounted for.

Every released flow row is accounted for



Full-release results retain support limits



Why another dataset name is not enough

- SCVIC: inspected clock fields did not qualify.
- DAPT: only 15 exfiltration rows and stage/time support constraints.
- DSRL: DAPT-derived synthetic construction limits independence.
- S-DAPT: withdrawn record; no qualified acquired replacement.

Two later executions were held out

- AIT release: eight testbeds; 3,465,342 source rows.
- Six executions train; two later executions evaluate.
- 2,397,158 training rows; 1,067,211 held-out rows.
- Three native classes; history is the optional evidence.

Policy outcomes on held-out runs

- Entropy: F1 0.9940; warnings 99.772%.
- Error-focused: F1 0.9462; warnings 99.727%.
- Benign alerts: 971.3 vs 14443.3.
- History acquired: 74.400% vs 2.450%.

Results differ across executions

- Wilson: F1 -0.0011; warnings +0.019 pp.
- Harrison: F1 -0.0821; warnings -0.102 pp.
- Original direction appears in 0 of six run/seed comparisons.
- Wilson has a small tradeoff in the opposite policy order.

Limits that bound the claim

- UNRAVELED is one campaign; AIT runs share a lab generator.
- Author labels do not independently verify stolen-file receipt.
- Completed-flow features do not establish early forecasting.
- Exact four-class replication and operational benefit remain open.

The deliverable is reviewable evidence

- Integrated manuscript with complete result appendices.
- Frozen protocols and complete source accounting.
- Saved models, per-row predictions and source archives.
- 622 full-release and 1,245 AIT audit checks passed.

The next test must retain the full task

- New workflow with all four native stage classes.
- Original role/history groups and collection constraints.
- Measure policy recurrence and label validity.
- Then test whether joint reports improve real review decisions.

Defensible conclusion

- Higher F1 can coexist with fewer exfiltration warnings.
- The original policy direction did not recur on AIT.
- Wilson shows a small tradeoff in the opposite policy order.
- Review scores, warnings and false alerts together.

Backup: original denominators

- Training: 147,087 benign; 12,362 other; 27 movement; 1,740 exfil.
- Calibration: 8,929 benign; 2,659 other; 0 movement; 1,331 exfil.
- Test: 192,193 benign; 12,424 other; 35 movement; 3,442 exfil.
- Temporal anchor: 104,051 rows; 18 movement-author-label rows.

Backup: warning accounting

- For true stage k : count benign, wrong-stage and correct-stage predictions.
- Warning recall = $1 - \text{benign predictions} / \text{stage-}k \text{ support}$.
- Exact recall = $\text{correct-stage predictions} / \text{stage-}k \text{ support}$.
- No support means unavailable recall; include the confusion matrix.

Backup: full-release class support

- net1011x: train [1466865, 0, 0, 0]
- net1012x: train [597531, 0, 0, 0]
- net1013x: train [837124, 12362, 129, 1740]
- net1014x: train [163270, 0, 0, 0]
- net1015x: train [144089, 0, 129, 0]
- netgw: train [2116400, 710, 13333, 6]

Backup: roles and prediction inputs

- Stable numeric current-flow fields; no absolute timestamps or raw endpoint IDs.
- Three destination-service indicators in both arms.
- Roles arm adds eight coarse endpoint-topology indicators.
- No future history or sensor ID enters either arm.

Backup: full-release fit uncertainty

- Extension: 300 boosting iterations; 15 leaves; learning rate 0.05; $L2 = 1$.
- Minimum child support 10; deterministic seed 20260927.
- One fit per sensor/arm; no new campaign confidence interval.
- Historical seed and capture omissions describe conditional sensitivity.

Backup: calibration choices

- net1011x: score → current; review → current
- net1012x: score → current; review → current
- net1013x: score → current; review → current
- net1014x: score → current; review → current
- net1015x: score → current; review → current
- netgw: score → current; review → current

Backup: closest primary sources

- Pendlebury et al. (2019), TESSERACT, USENIX Security.
- Bilot et al. (2025), Sometimes Simpler Is Better, USENIX Security.
- Guerra et al. (2026), benchmark/protocol study, preprint v3.
- Myneni et al. (2023), UNRAVELED source release.

Backup: likely committee questions

- Novelty: controlled evidence and reproducible review.
- Full data: both source releases audited; new fits uncapped.
- External test: adapted ALT; original direction unreplicated.
- Exact replication and operational benefit remain future work.

Backup: complete fitting inventory

- Historical core: 141 LightGBM fitting jobs.
- Technique-policy supplement: two Ridge selector fits.
- Full-release extension: 12 sensor/arm fitting jobs.
- AIT extension: 36 classifier and six selector fits.

Backup: executable calibration review

- Score-only choice: select higher calibration macro-F1.
- Review choice: also constrain stage-warning loss and benign FPR.
- Freeze the choice, then evaluate the same later test rows.
- All six views lack at least one attack stage in calibration.

Backup: the adapted AIT method

- Three classes: benign, other attack, exfiltration.
- Optional evidence: strictly earlier flow history.
- Selectors compare entropy and weighted error.
- Fixed design; three seeds; no test-set tuning.

Backup: AIT interpretation limits

- External-source evaluation is now completed.
- The original four-class experiment remains distinct.
- Native exfiltration labels use publisher rules.
- Completed-flow results do not establish early warning.